# Route Wise — Market Requirements Document (MRD)

**DOC:** MRD-v2.6-GOLD | **TEAM:** Cyber Knight
**STATUS:** Approved & Verified (99 Tests)
**AUDIENCE:** Evaluators, Mentors & Judges

**The 30-Second Elevator Pitch:** Most navigation apps blindly suggest the shortest route, funneling traffic into severe junction gridlock. **Route Wise (Team Cyber Knight)** is an intelligent traffic advisory platform, live glassmorphic dashboard, and installable PWA. It evaluates multiple driving alternatives, penalizes junction bottlenecks using Adjusted ETA with an anti-flapping switch threshold, predicts 15-minute trends via public **ARIMA (1,0,1)** ML, correlates crowd-sourced incidents with TomTom speed-drop telemetry via 2-vote consensus, provides municipal authority recurrent hotspot triage, and guarantees corridor retention during 1-tap Google Maps handoff.

## 1. Problem Framing & 5-Why Root Cause Analysis

**Validated Problem Statement:** Daily commuters and couriers face 15-to-30-minute delays at junction bottlenecks in Tier-1/2 Indian cities (e.g. Tiruchirappalli), causing missed SLAs, excess fuel burn, and stress. Standard GPS algorithms over-weight physical distance and lack transparent trade-off explanations. Therefore, there is an urgent need for an explainable routing system that penalizes junction gridlock and validates crowd alerts.

| 5-Why Diagnostic Root Cause Tree | Analysis & Evidence |
|---|---|
| **1. Why delayed?** | Trapped in bumper-to-bumper junction gridlock during peak hours. |
| **2. Why enter?** | Standard GPS displays the junction route as the shortest path in kilometers. |
| **3. Why shortest?** | Incumbent APIs use opaque black-box weights without customizable junction delay penalties. |
| **4. Why no detour?** | Commuters receive no side-by-side trade-off explanations (e.g. '+2.5 km but 16m faster'). |
| **5. Root Cause** | **Lack of an explainable engine combining live delay, ML forecasting, and verified crowd alerts.** |

**Real-World Validation: Trichy Peak Rush Hour (CBS ■ Thillai Nagar across Cantonment)**

| Navigation System | Distance | Junction Delay | Trip Time | Driver & Courier Outcome |
|---|---|---|---|---|
| **Standard Navigation App (Shortest distance default)** | 10.0 km (Shortest) | **+18 mins** (HPO bottleneck) | **42 mins** | Late delivery, customer SLA breach, high fuel wasted idling in gridlock. |
| **Route Wise (Cyber Knight) (Adjusted ETA + ARIMA ML)** | 12.5 km (+2.5 km) | **+2 mins** (Smooth flowing) | **26 mins (16m faster!)** | On-time delivery, clear trade-off rationale, fuel-efficient constant speed corridor. |

## 2. Market Opportunity & Sizing (Assumptions Explicitly Marked)

| | | |
|---|---|---|
| **TAM: $14.2 Billion**<br>Global GIS, Navigation & Fleet Logistics API Market (CAGR 12.4%)<br>**Assumption: Worldwide logistics transition to API-driven dispatch.* | **SAM: $2.8 Billion**<br>Urban Last-Mile Delivery & Commuter Routing Software Platforms<br>**Assumption: 20% of urban fleets adopt dynamic junction-avoidance software.* | **SOM: $350 Million**<br>Tier-1 & Tier-2 Indian Logistics Startups, Fleets & Commuters<br>**Assumption: Initial focus on South Indian freight hubs (Trichy, Coimbatore, Chennai).* |

# ROUTE WISE (CYBER KNIGHT) — MARKET REQUIREMENTS (PAGE 2)

## 3. Target Customer Personas

| Persona | Role & Context | Pain Points & Unmet Needs | Route Wise Value Delivered |
|---|---|---|---|
| **Karthik (26)**<br>**Commuter / Courier** | On-demand delivery courier & daily 2W commuter in Trichy. | Stuck at unannounced signal failures; misses strict 8 deliveries/hr SLA. | Trichy junction presets, 0ms autocomplete, 1-tap Google Maps handoff. |
| **Rajesh (38)**<br>**Fleet Dispatcher** | Oversees 45 regional light commercial delivery vans across TN. | Fuel budget overruns by 14% due to idling; GPS apps flap routes erratically. | Adjusted ETA with anti-flapping threshold; 60s live auto-refresh. |
| **Insp. Sundaram (48)**<br>**Traffic Authority** | Municipal traffic police & smart city urban planner. | Flood of citizen complaints without location data; no recurrence tracking. | Authority Hotspot Dashboard with 7-day spatial recurrence audit log. |

## 4. Commuter User Journey: Current Gridlock vs Proposed Route Wise Flow

| Journey Stage | Current Commuter Friction (Without Route Wise) | Proposed Route Wise Experience | Impact |
|---|---|---|---|
| **1. Trip Planning** | Enters destination; app selects single shortest path with zero trade-off info. | Compares up to 3 candidate driving routes with transparent Adjusted ETA. | **100% transparency** |
| **2. Junction Triage** | Enters bottleneck blindly; signal breakdown discovered only after arrival. | 15-minute ARIMA ML trend and verified community incident markers visible. | **Proactive avoidance** |
| **3. Navigation** | Opening Google Maps recalculates and reverts back to congested main arterial. | Decision-anchor waypoints (≤ 3 waypoints) preserve chosen corridor. | **Corridor fidelity** |

## 5. Competitive Landscape & Value Proposition

| Feature / Capability | Google Maps API | Mapbox Navigation | Waze Platform | Route Wise (Cyber Knight) |
|---|---|---|---|---|
| **Configurable Scoring Weights** | No (Fixed defaults) | Partial (Profiles) | No (Fixed defaults) | **Yes — Adjusted ETA formula** |
| **Transparent 'Why' Rationale** | No (Opaque path) | No (No explanation) | No (No explanation) | **Yes — Comparative plain-English tags** |
| **15-Min ML Traffic Forecast** | No (Live snapshot only) | No (Historical only) | No (Live alerts only) | **Yes — Public ARIMA (1,0,1) model** |
| **Anti-Flapping Route Threshold** | No (Erratic flipping) | No (Flips on 1s delta) | Partial (Basic dampening) | **Yes — max(180s, 10% ETA)** |
| **Corridor-Preserving GMaps Handoff** | N/A | No (Raw coords only) | N/A | **Yes — Decision-anchor waypoints** |
| **Incident Verification Consensus** | Proprietary / Closed | No incident reporting | Unverified noise | **Yes — 2-Vote + speed drop (>30%)** |
| **Offline Demo Fail-Safe** | No (Fails on network drop) | No (Paid offline packs) | No (Network required) | **Yes — Built-in dictionary + PWA** |

## 6. Business Model & Strategic Risk Mitigations

| Tiered Monetization Model | Strategic Risk & Verified Mitigation Matrix |
|---|---|
| • **Developer Free ($0/mo):** 1,000 queries/mo, 60 req/min limit, full dashboard access, offline mock mode.<br>• **Growth Fleet ($0.003/query):** Up to 250k queries/mo, custom scoring profiles, 60s live refresh API, ARIMA ML forecast.<br>• **Municipal / Enterprise SLA:** Unlimited queries, Authority Hotspot Dashboard, 99.9% uptime SLA. | • **API Quota / Network Outage:** Local coordinate dictionary ensures 100% demo uptime.<br>• **Spam Crowd Reports:** 2-vote peer consensus + TomTom speed-drop (>30%) correlation.<br>• **Route Reversion on Handoff:** Decision-anchor waypoints (≤ 3 waypoints) preserve corridor.<br>• **Database Contention:** Native Node 24 SQLite in Write-Ahead Logging (`WAL`) mode. |